A general solver for electrochemical transport-reaction systems

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THE CLASS OF PROBLEM

One framework, many systems

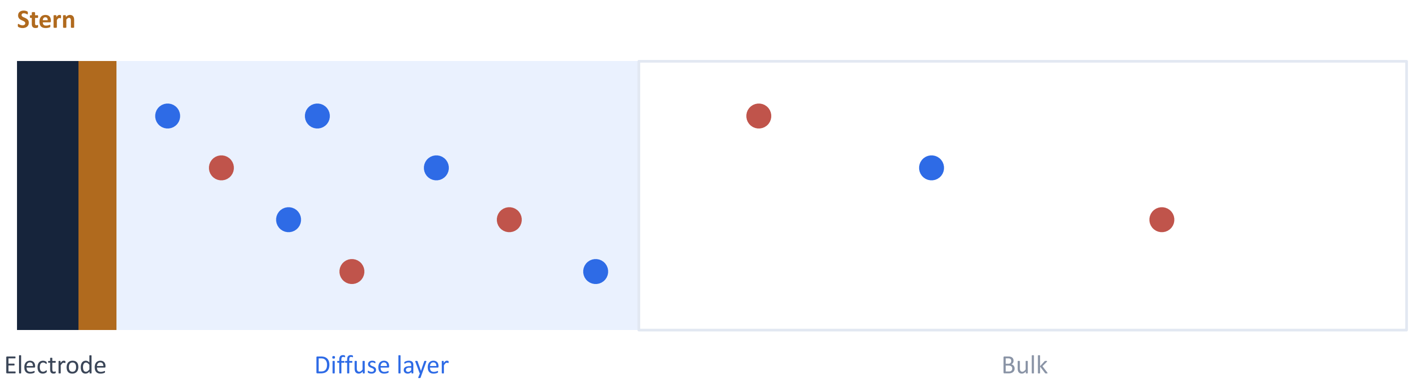
Migration–diffusion–reaction with a structured double layer — the same mathematics across electrochemistry.

ORR

CO_2RR

Electrodeposition

Corrosion



We'll demonstrate the workflow on ORR.

THE IDEA

Two halves of the problem

Model specification

the electrochemistry

- Species
- Reactions & stoichiometry
- PDE terms
- Boundary conditions

→

Numerical machinery

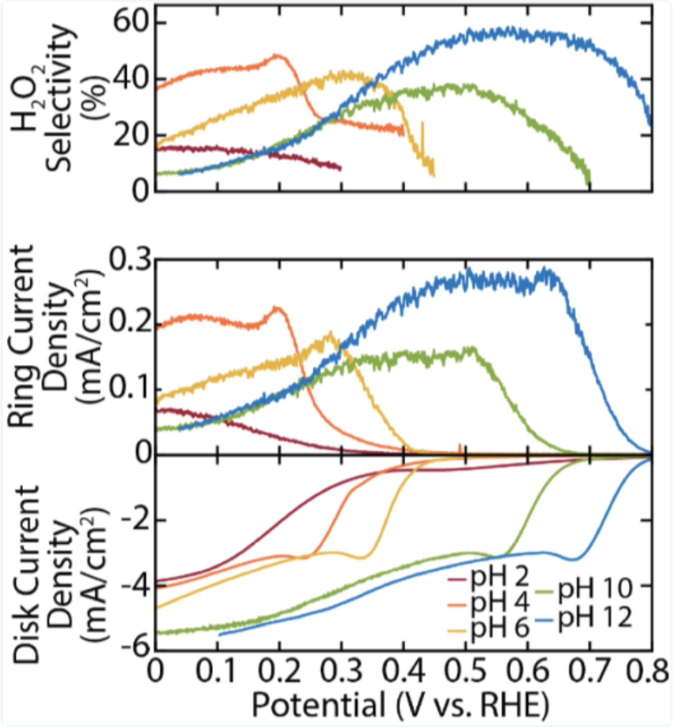
the solver's job

- Formulation & log-space
- Meshing (1D / 2D)
- Continuation & recovery
- Robust convergence

Proof of concept: ORR selectivity

The system:

- Parallel $2e^-$ (0.695 V) + $4e^-$ (1.23 V)
- K^+ / SO_4^{2-} electrolyte
- Finite-size ions (Bikerman)
- Stern + Butler–Volmer + bulk BCs



Target: Ruggiero / Seitz–Mangan, pH 2–12

What you specify

Governing Equations

PNP in the diffuse layer

$$\frac{\partial c_i}{\partial t} = \nabla\cdot\left(D_i\left[\nabla c_i + \frac{z_i F}{RT}\, c_i\, \nabla\varphi\right]\right) \qquad -\nabla\cdot(\varepsilon\,\nabla\varphi) = \sum_i z_i F\, c_i$$

Butler–Volmer flux

$$R_r = k_{0,r}\left[c_O\, e^{-\alpha_r F\eta_r/RT} - c_R\, e^{(1-\alpha_r)F\eta_r/RT}\right], \qquad \eta_r = \varphi_m - \varphi_s - E^{\circ}_{,r}$$

Steric correction

$$\mu_i^{\text{steric}} = -\ln\theta, \qquad \theta = 1 - \sum_i a_i c_i$$

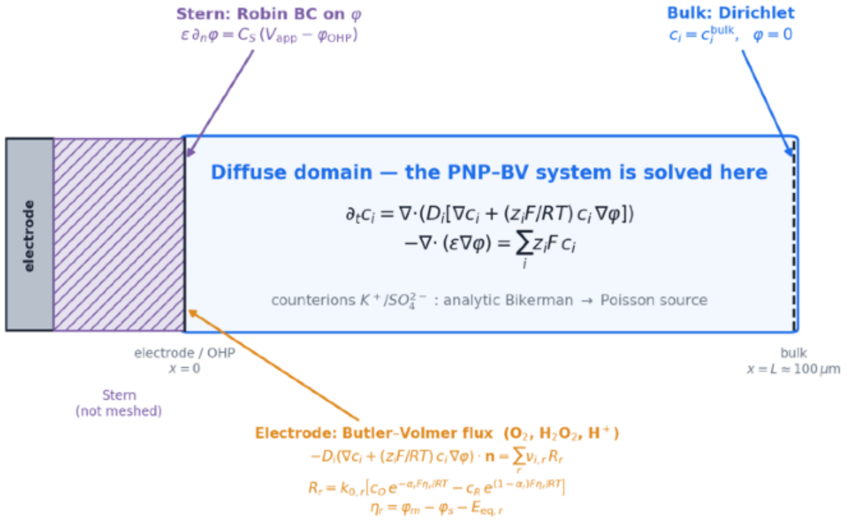
Stern layer

$$\varepsilon\,\partial_n\varphi = C_S\,(V_{\text{app}} - \varphi_{\text{OHP}}), \qquad C_S = 0.20\ \text{F/m}^2$$

What you specify

Boundary Conditions

Dirichlet | Butler–Volmer | Robin | No-flux



Mix and match per species, per boundary.

What you specify

Reactions

$O_2 + 2\,H^+ + 2\,e^- \rightarrow H_2O_2$

$E° = 0.695$ V · $2e^-$

$O_2 + 4\,H^+ + 4\,e^- \rightarrow 2\,H_2O$

$E° = 1.23$ V · $4e^-$

Declare stoichiometry · n_e · $E°$ · α · k_0 → log-rate Butler–Volmer (parallel or sequential)

Why it's numerically hard

- **Concentrations span ~10 orders of magnitude** across the domain.
- **The potential varies sharply** right at the Stern layer.
- **The exponential Butler–Volmer term** drives steep buildup in c and ϕ at the boundary.
- **Diffusion and electromigration** act on very different timescales — the system is stiff.

$$c_i \approx c_i^{\text{bulk}} \exp\left(-z_i F \varphi / RT \right)$$

How we solve it

1 **Solve μ = ln c + z·ϕ** — mirrors the no-flux analytic solution.

$$\mu_H = u_H + \frac{z_H F}{RT}\varphi$$

2 **Log-rate Butler–Volmer** — build the rate in log-space, exponentiate once (avoids overflow).

$$R_r = k_{0,r}\left[c_O\, e^{-\alpha_r F\eta_r/RT} - c_R\, e^{(1-\alpha_r)F\eta_r/RT}\right]. \longrightarrow$$

$$\log r_r^{(\mathrm{cat})} = \ln k_{0,r} + u_O - \alpha_r \frac{F\eta_r}{RT}, \qquad r_r^{(\mathrm{cat})} = \exp\big(\log r_r^{(\mathrm{cat})}\big),$$

$$\log r_r^{(\mathrm{anod})} = \ln k_{0,r} + u_R + (1-\alpha_r)\frac{F\eta_r}{RT}, \qquad r_r^{(\mathrm{anod})} = \exp\big(\log r_r^{(\mathrm{anod})}\big),$$

$$R_r = r_r^{(\mathrm{cat})} - r_r^{(\mathrm{anod})}, \qquad \eta_r = \varphi_m - \varphi_s - E^{\circ}_{,r},$$

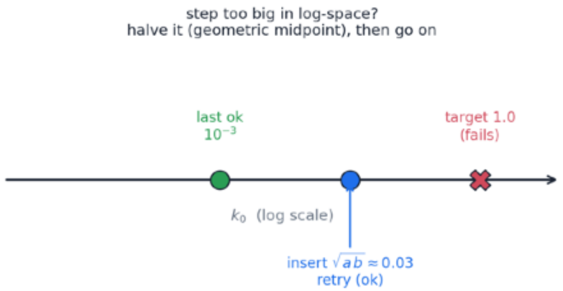
3 **Analytic closure for non-reactive ions.**

$$c_b(x) = c_b^{\mathrm{bulk}}\frac{\left(1 - \sum_{j\in\mathrm{dyn}} a_j c_j\right) e^{-z_b F\varphi/RT}}{\theta_b + a_b\, c_b^{\mathrm{bulk}}\, e^{-z_b F\varphi/RT}}, \qquad \theta_b = 1 - \sum_j a_j c_j^{\mathrm{bulk}} - a_b c_b^{\mathrm{bulk}}$$

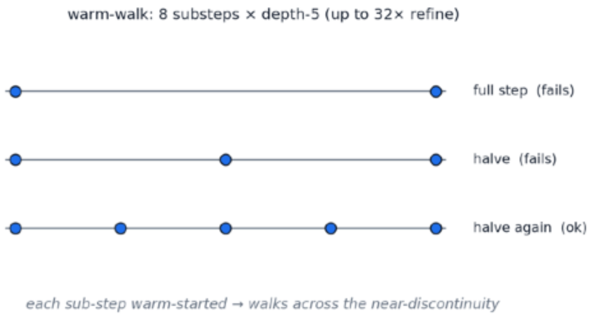
4 **Continuation** — never solve the hard problem cold (next slide).

Continuation: deform easy → hard

- **Reaction rate k_0:** ramp from ~10^{-12}× physical up to physical, warm-starting each step.
- **Stern capacitance:** ramp up (stiffer as C_S grows).
- **Voltage walk:** converge near the equilibrium voltage, then step out across the window.
- **Recursive bisection:** insert intermediate steps when one fails.

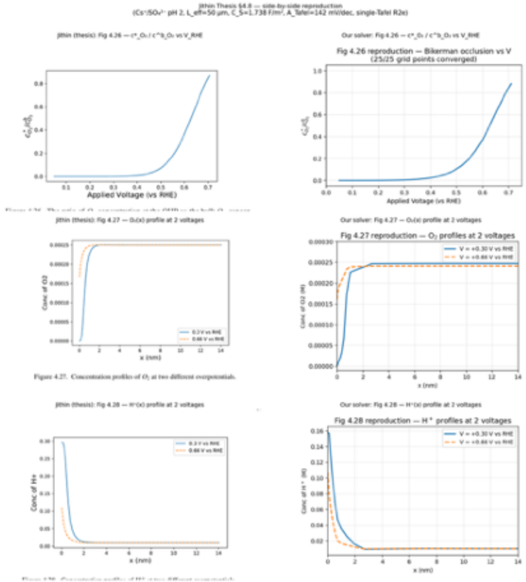


k0 laddering

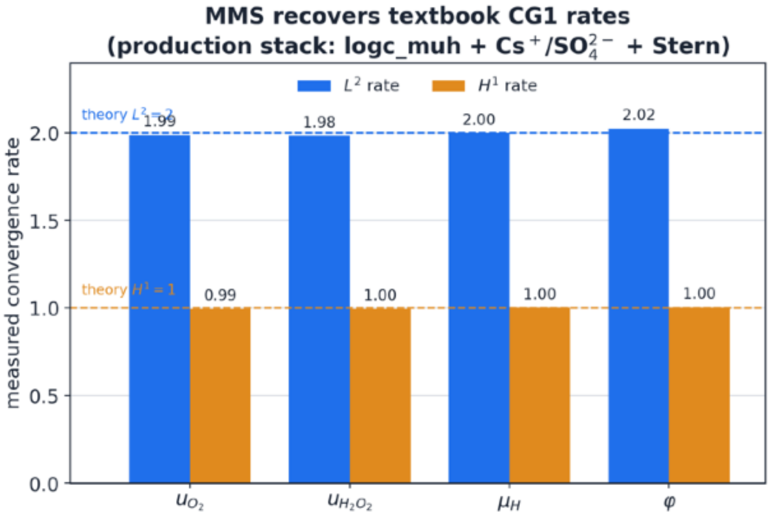


Warm walking from an anchor voltage

What's proven — and what's left



Reproduces the prior model (~5%)



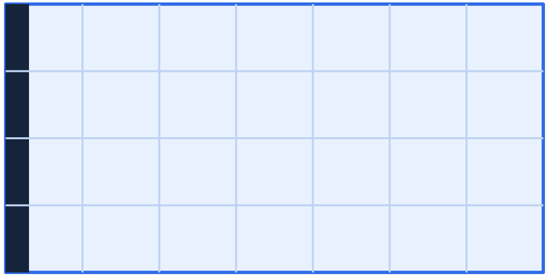
MMS → textbook convergence

The remaining gap to experiment match is model selection, not solver convergence.

1D and 2D

1D

2D



Where it's headed: a Python package

```python
import pnpbv as pb

O2 = pb.Species("O2", z=0,  D=2.1e-9, c_bulk=1.2e-3)
H  = pb.Species("H+", z=+1, D=9.3e-9, c_bulk=1e-4)
K  = pb.Species("K+", z=+1, c_bulk=0.2,
                analytic=True)          # Boltzmann

r2e = pb.BVReaction("ORR_2e",
         reactants={O2:1, H:2}, products={H2O2:1},
         n_e=2, E_eq=0.695)

prob = pb.Problem(
    species=[O2, H2O2, H, K, SO4],
    reactions=[r2e, r4e],
    electrode=pb.SternElectrode(C_stern=0.20))

sol = pb.Solver(prob).solve_grid(v_rhe=(-0.4, 0.8))
```

Declare the spec.

It handles meshing, formulation, solving, and recovery.

Planned: hyperparameter autotuning.

Design sketch — not yet built

Adding terms

```python
from pnpbv.formulations import pnp_steric
from ufl import grad, inner

# default PNP + migration + Bikerman steric:
res   = pnp_steric(prob)
c, w = res.fields, res.tests    # conc + tests
dx    = res.dx                  # 2D bulk measure

# add terms the menu doesn't cover:
# peroxide decomposition:  H2O2 -> 1/2 O2 + H2O
res.add( -k_d      * c["H2O2"] * w["H2O2"] * dx )
res.add( +0.5*k_d * c["H2O2"] * w["O2"]   * dx )
# example convection term
res.add( inner(u, grad(c["O2"])) * w["O2"] * dx )

# same solver
sol = pb.Solver.from_weak_form(res).solve_grid(...)
```

Import the default weak form residual,
append terms as plain UFL.

The same solver, continuation, and recovery.

Design sketch — not yet built

Going forward

1. **Close the experiment gap by model selection**

2. **Broaden the menu of options & add autotuning**
   more terms, BCs, and reactions; let the solver pick its own settings

3. **Lock the API**
   a reusable predictive simulator for systems in this family